Fowards

- In this lecture, we begin our study of derivatives
- We will start with *forwards*
- Standardized implementations of forwards are called *futures*
- Henceforth, I will refer to everything as a "forward"
- We will then use forwards to study
  (1) swaps
  (2) exchange rates

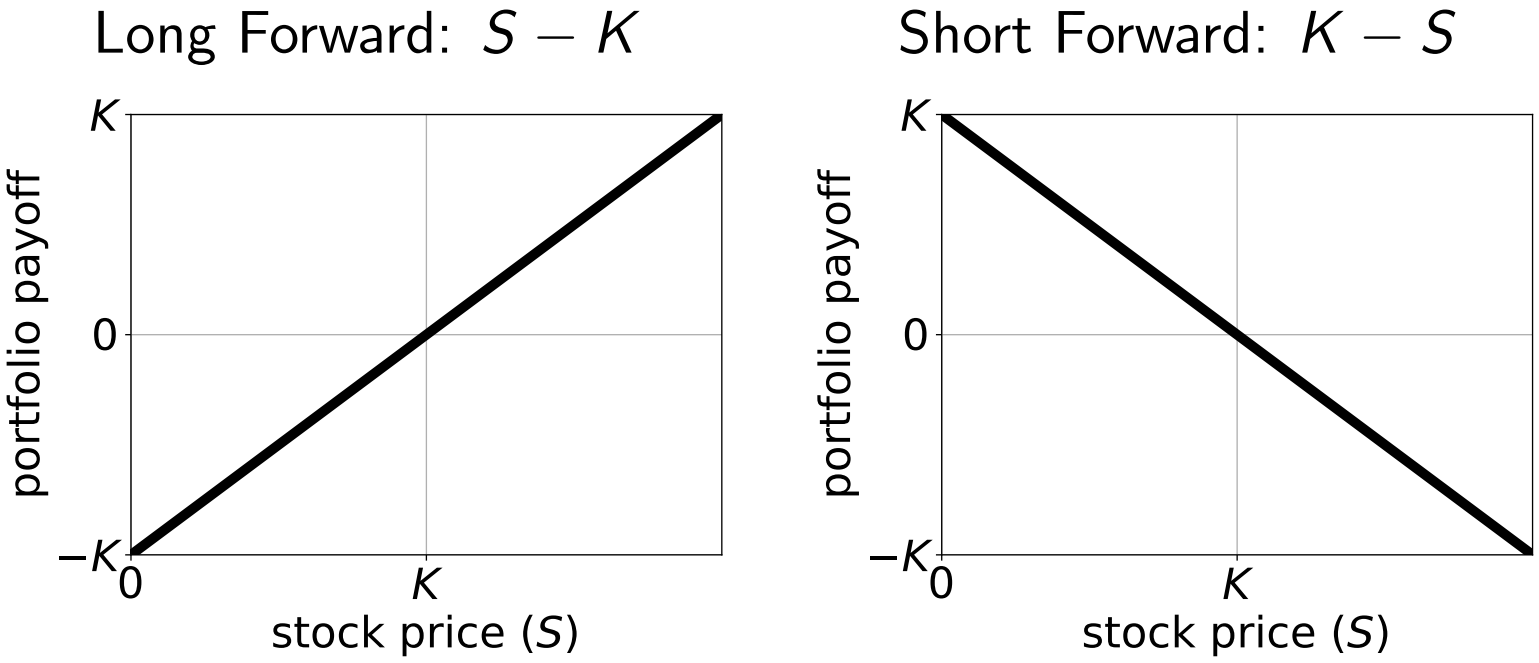
Vocabulary

- A *derivative* is a security whose payoff depend on the payoff of other securities
- A *forward* is a contract that requires future delivery of asset at agreed-upon price
- *Cash settlement* means delivering the cash value of the underlying asset (rather than the asset itself) to satisfy the contract
- The *underlying* security (stock, bond, currency, commodity) on which the forward is written
- The *future price* is the agreed-upon price to be paid on a futures contract at maturity
- The *spot price* is the actual market price of the security

Mechanics

- If you are *long* a forward, then you commit to *sell* the security, while if you are short, you commit to *buy* the security
- With cash settlement, the party who is short does not have to literally sell the underlying security to her counterparty; rather, she can just pay her counterparty the difference between the spot price and the future price in cash
- Since positions are settled at maturity, there is no initial cash flow: everything happens "in the future"

Forward Payoffs



- The trader in the long position is said to "buy" a contract; the short-side trader "sells" a contract.
- Basis: difference between futures price and spot price $K - S$

Forward Payoffs: Problem 1

Problem. Suppose Apple (AAPL) trades for 270 USD. James Simons (Renaissance Technologies) is going to write a 3-year

Corporate Applications: a Vignette

- Consider an oil company and an airline.
- How much will the price of oil be in the future, when they are ready to sell the oil they produced?
- The revenue of this company depends on oil prices, which are extremely volatile.
- The airline has a problem that mirrors that of the oil production company.
- How can forwards help these two firms?

Corporate Applications: a Vignette

- The two firms can enter into an agreement today that specifies the price at which the oil producer sells the oil to the other company in the future.
- Such an agreement will reduce the source of risk for both parties!
- The oil producer delivers oil at a price agreed upon now, regardless of the market price at the time when oil is produced.
- This requires that both parties be willing to lock in the price to be paid or received for delivery of the commodity.
- This can be achieved using forwards.

Corporate Applications

Southwest Airlines gains advantage by hedging on long-term oil contracts

By **JEFF BAILEY** NOV. 28, 2007

CHICAGO — Southwest Airlines, in danger for much of this year of losing its quirky dominance over the U.S. domestic airline industry, could soon be standing, once again, head-and-shoulders above its competition.

Better service? Happier and more productive workers?

Not this time. The reason for Southwest's rapidly increasing advantage over other big airlines is much simpler: It loaded up years ago on hedges against higher fuel prices. And with oil trading above $90 a barrel, most of the rest of the airline industry is facing a huge run-up in costs, and Southwest is not.

Southwest owns long-term contracts to buy most of its fuel at the equivalent of $51-a-barrel oil through 2009. The value of those hedges soared as oil raced above the $90-a-barrel mark and they are now worth more than $2 billion.

Those gains would mostly be realized during the next two years.

Other major airlines passed on buying all but the shortest-term insurance against high fuel prices, giving Southwest executives a mild case of schadenfreude.

"It's true," said Scott Topping, treasurer at Southwest and keeper of the hedges. "We're not sure what to root for," in terms of oil prices. Southwest is hurt, too, by higher fuel prices, but far less than competitors, giving the carrier a distinct

Spot-Futures Parity

- Having looked at the mechanics and applications of forwards, we know consider their pricing
- T is the term, S_0 is the spot price on date 0, S_T is the spot price on date T, and K is the future price
- Spot-Futures Parity $K \overset{\text{"should"}}{=} S_0(1+r_F)^T$

Security	Q	CF_0	CF_T
Forward	1	0	$S_T - K$
Stock	1	$-S_0$	S_T
Risk-Free	$-K/(1+r_F)^T$	$K/(1+r_F)^T$	$-K$
		$-S_0 + K/(1+r_F)^T$	$S_T - K$

Spot-Futures Parity: Hedging Problem

Problem: You're in sales at Goldman Sachs. A client wants a five-year forward on 1 m USDof Apple, which currently sells for 270 USD. The annually compounded risk-free rate is .25%. What should the price K be? How do you hedge the position?
Solution: The futures price is

$$K = S_0(1 + r_F)^T = (1 \text{ m USD})(1 + .0025)^5 = 1.0126 \text{ m USD} \quad (1)$$

This means that in five years, Goldman Sachs is obligated to sell the client (1 m USD/270 USD) = shares of Apple for a total of 1.0136 m USD. In order to hedge this transaction, Goldman Sachs can buy

Spot-Futures Parity: Margin Problem

Motivation. Because both parties to a futures contract are exposed to losses, both must post margins.
Problem. Suppose that you buy a one-year forward on 10k shares of SPY, which currently sells for 275 USD. The risk-free rate is .25%. You are required to post an initial margin of 50%.

1. What should the forward price be?
2. How much margin (USD) must you post?
3. If SPY drops by 5% tomorrow, what will the new forward price be (approximately)?
4. How much margin do you have left?
5. What is your return?

Spot-Futures Parity: Margin Problem

Solution.

1. The forward price (per share) is

$$K = S_0(1 + r_F)^T = (275 \text{ USD})(1 + .0025) = 275.6875 \text{ USD} \quad (2)$$

2. You must post
3. Suppose that SPY drops by 5% tomorrow: $S_0 = (95\%)(275 \text{ USD}) = 261.25 \text{ USD}$. If ignoring the 1 day that has past,

$$K = S_0(1 + r_F)^T = (261.25 \text{ USD})(1 + .0025) = 261.9031 \text{ USD} \quad (3)$$

4. How much margin do you have left?
5. What is your return?

Pricing Existing Forwards

- Forwards, like all of the derivatives we will study, can be sold in a secondary market
- Suppose you go long a forward that expires in <u>two</u> years.
- Step forward one year.
- Consider a new forward that expires in <u>one</u> year.
- How much could you sell your old forward for (say f_0)?
- This is the problem of pricing an existing forward

Pricing Existing Forwards

How do you price anything? Build a replicating portfolio!

Security	CF_1	CF_2
Old Forward (K_0)	$-f_0$	$S_2 - K_0$
Replicating Portfolio		
New Forward (K_1)	0	$S_2 - K_1$
Risk Free	$-(K_1 - K_0)/(1 + r_F)$	$K_1 - K_0$
	$-(K_1 - K_0)/(1 + r_F)$	$S_2 - K_0$

$$\boxed{f_0 \overset{\text{“should”}}{=} \frac{K_1 - K_0}{1 + r_F}} \tag{4}$$

Pricing Existing Forwards: Problem 1

Problem. Suppose that the stock price was $S_0 = 100$ USD a year ago and the stock price is $S_1 = 101.50$ USD today. What is the future price of a two-year forward written a year ago? Suppose that the annually compounded risk-free rate is $r_F = 1.5\%$

Solution.

- The old and new forward prices are

$$K_0 = S_0(1 + r_F)^2 = 121 \text{ USD} \tag{5}$$

$$K_1 = S_1(1 + r_F) = 126 \text{ USD} \tag{6}$$

- The old forward trades now trades for

$$f_0 = \frac{K_1 - K_0}{1 + r_F} = \frac{126 \text{ USD} - 121 \text{ USD}}{1 + .1} = 4.54 \text{ USD} \tag{7}$$

Pricing Existing Forwards: Problem 2

Problem. $S_0 = 100$ USD, $S_1 = 110$ USD, $r = 10\%$

Solution.

- The old and new forward prices are

$$K_0 = S_0(1 + r_F)^2 = 121 \text{ USD} \tag{8}$$

$$K_1 = S_1(1 + r_F) = 121 \text{ USD} \tag{9}$$

- The old forward trades for

$$f_0 = \frac{K_1 - K_0}{1 + r_F} = \frac{121 \text{ USD} - 121 \text{ USD}}{1 + .1} = 0 \text{ USD} \tag{10}$$

Application I: Swaps

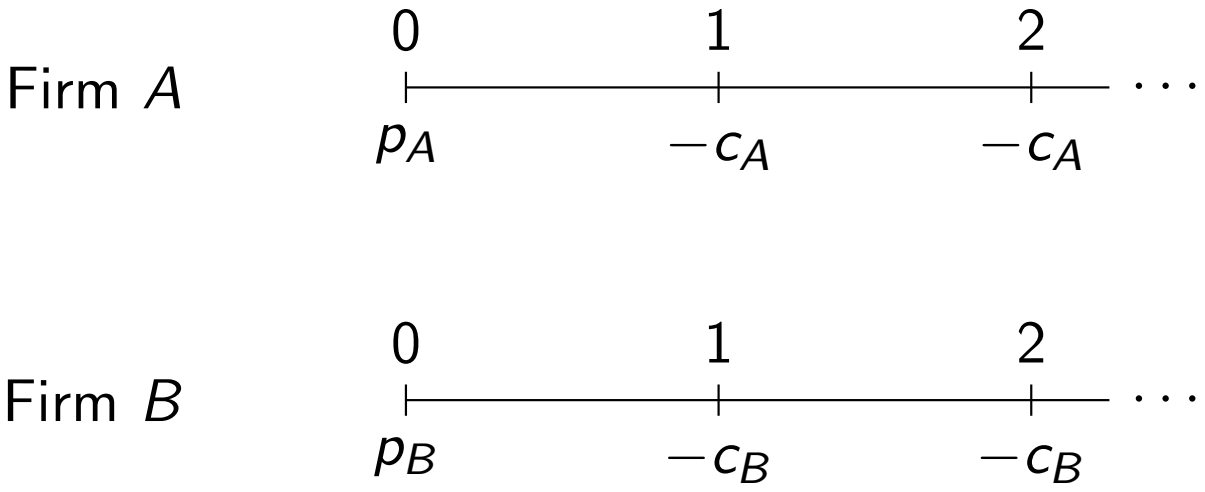
- A swap is a contract in which parties "swap" cash-flows.
- Party A has cash flows $c_1^A, c_2^A, \ldots$
- Party B has cash flows $c_1^B, c_2^B, \ldots$
- A and B swap: B pays A $c_1^B, c_2^B, \ldots$ and A pays B $c_1^A, c_2^A, \ldots$
- Swaps are technically portfolios of forwards

Swaps: IBM and the World Bank

- Salomon Brothers wrote the first swap in 1981
- The World Bank is an international organization that borrows from developed countries and lends to developing countries
- In 1981, Swiss debt was cheap, so the World Bank wanted to borrow there, but...
- Switzerland limited how much the World Bank could borrow
- IBM had already borrowed in Switzerland and wanted dollars
- Salomon proposed "swapping" cash flows
- The World Bank borrowed dollars
- IBM paid the World Bank's dollar obligations
- The World Bank paid IBM's franc obligations
- It was as if the World Bank borrowed in francs and IBM borrowed in dollars

Swaps: Pricing

- A borrows p_A, pays coupon c_A each period
- B borrows p_B, pays coupon c_B each period



- If the loans are priced correctly, then for rates r_A and r_B,

$$p_A = \frac{c_A}{1 + r_F} + \frac{c_A}{(1 + r_F)^2} + \cdots \tag{11}$$

$$p_B = \frac{c_B}{1 + r_F} + \frac{c_B}{(1 + r_F)^2} + \cdots \tag{12}$$

Interest Rate Swaps

w_0 is the price of the swap

Security	CF_0	CF_1	CF_2
		Firm A	
Loan	$+p_A$	$-c_A$	$-c_A$
Swap	$-w_0$	$c_A - c_B$	$c_A - c_B$
	$p_A - w_0$	$-c_B$	$-c_B$
		Firm B	
Loan	$+p_B$	$-c_B$	$-c_B$
Swap	$+w_0$	$-(c_A - c_B)$	$-(c_A - c_B)$
	$p_B + w_0$	$-c_A$	$-c_A$

Interest Rate Swaps: Pricing

w_0 is the price of the swap

Security	CF_0	CF_1	CF_2
Swap	$-w_0$	$c_A - c_B$	$c_A - c_B$
	Replicating Portfolio		
Loan A	$-p_A$	$+c_A$	$+c_A$
Loan B	$+p_B$	$-c_B$	$-c_B$
	$-p_A + p_B$	$c_A - c_B$	$c_A - c_B$

$$\boxed{w_0 \overset{\text{“should”}}{=} p_A - p_B} \tag{13}$$

Interest Rate Swaps: Problem

Suppose that the yield on a one-year bond is 5% (${}_0y_1 = 5\%$) and the yield on a two-year bond is 10% (${}_0y_2 = 10\%$). What should the swap trade for (w_0)?

Application II: Exchange Rates

- International finance is a large and rich topic
- Today, we're going to price currencies
- We're going to study a couple of practical examples
- Currencies are like tacos
- I go to 3 Amigos and buy tacos, I go to Chase and buy pesos
- In both cases, I pay with dollars
- Let E denote the price of foreign currency
- E.g. a Mexican peso sells for $E = .04$ USD

Interest Rate Parity

- Two countries, domestic (D) and foreign (F), r_D is the risk-free rate in D, r_F in F, E_0 is the spot price of F's currency, and K_0 is the price of a forward on F's currency
- *Interest Rate Parity* occurs when

$$\boxed{1 + r_D = (1 + r_F) \cdot \frac{K_0}{E_0}} \tag{14}$$

- Put differently, you should not be able to make more money on a Euro denominated risk-free asset (e.g. German bunds) than on a Dollar denominated risk-free asset (e.g. US Treasuries)

Interest Rate Parity: Problem

Problem. 1 AUD costs $E_0 = .64$ USD. The yields on 1-year Australianand US government bonds are .245% and .3% respectively. What should the 1-year forward price of 1 AUD be? If the 1-year forward price of 1 AUD is $K_0 = .6402$ USD, what arbitrage opportunity is available to you?

Solution. Let's first compute what the forward price should be under Interest Rate Parity:

$$K_0 = \frac{1 + r_D}{1 + r_F} \cdot E_0 = \frac{1 + .00300}{1 + .00245}(.64 \text{ USD}) = .6404 \text{ USD} \tag{15}$$

As you can see, the 1-year forward price is .6402 USD is less than what it should be under Interest Rate Parity, so we want to <u>borrow</u> AUD and <u>lend</u> USD.

Interest Rate Parity: Problem

Solution (continued). To exploit the arbitrage opportunity, use the following procedure.

Today,

1. borrow 1 m AUD,
2. buy 1 m AUD $\times$ (.64 USD / AUD) $=$ .64 m USD
3. buy .64 m USD of US government bonds
4. sell a 1-year forward for .6402 USD

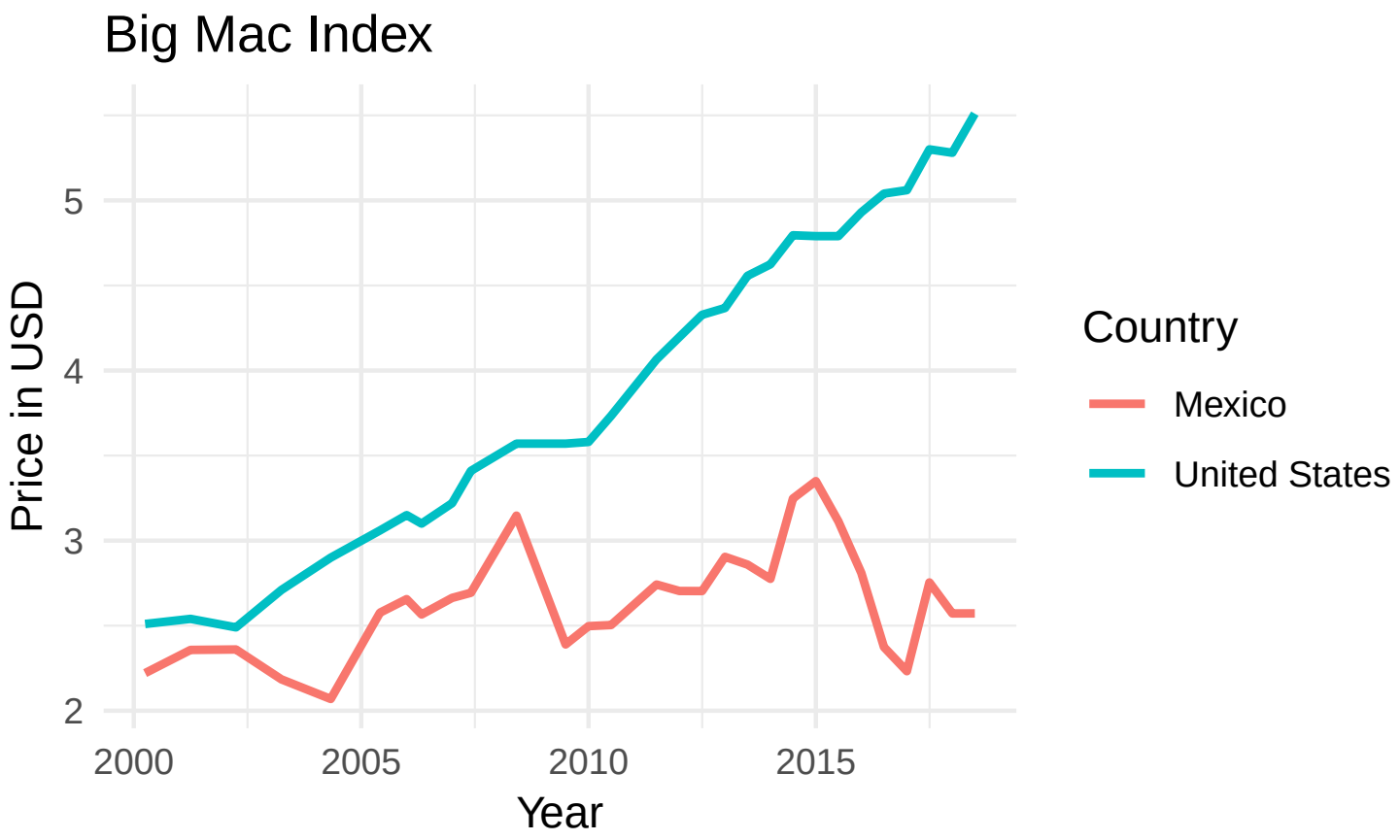
In one year,

1. collect $(1 + .003) \times$ (.64 m USD) $=$ .6419 m USD
2. .6419 m USD /(.6402 USD / AUD) $=$ 1.0027 m AUD
3. repay 1 m AUD $\times$ $(1 + .00245) = 1.0025$ m AUD
4. you have collected a profit of .0002 m AUD (about 200 AUD)

Interest Rate Parity vs. Purchasing Power Parity

- Can we apply the same logic to other assets?
- *Purchasing power parity* refers to the ability to buy the same basket of goods/services around the world for the same price
- A taco in the US sells for 3 USD, a taco in Mexico sells for 20 MXN, a peso sells for .05 USD. What arbitrage opportunity is available to you?
- Buy 20 MXN for 1 USD, buy 1 taco in Mexico for 20 MXN, sell 1 taco in the US for 3 USD. You've made 2 USD.
- Money can be moved around the world (almost) costlessly. Tacos cannot.

The Economist's Big-Mac Index



Although interest rate partity tends to hold in the data, does not. This is call the Penn effect and it is due to (1) limits to arbitrage and (2) differences in economic development.

Problems

1. Forward Payoffs: Problem 1
2. Spot-Futures Parity: Hedging Problem
3. Spot-Futures Parity: Margin Problem
4. Pricing Existing Forwards: Problem 1
5. Pricing Existing Forwards: Problem 2
6. Interest Rate Swaps: Problem
7. Interest Rate Parity: Problem